

# Mingyang (Yang) Liu

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Imperial College London

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On the 2026–27 academic job market

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## Research Fields

Empirical asset pricing (cross-sectional return prediction, characteristic-based factor models); machine learning and big data in finance (regularised and latent-factor estimators on large characteristic panels); financial econometrics (identification and inference in high-dimensional factor models); portfolio choice and risk management.

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## Positions

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| Jun 2026–present | <p><b>Visiting Researcher</b>, Imperial Centre of Excellence in Quantitative Finance, Department of Finance, Imperial Business School, Imperial College London</p> <p>Host: Prof. Robert Kosowski, Co-Director of the Centre</p> <p>Research agenda for the visit: economically constrained machine learning for cross-sectional asset pricing and systematic investment.</p> <ul style="list-style-type: none"><li>• Machine learning and high-dimensional data: extending the No-Arbitrage IPCA framework of the thesis to regime-switching factor models in which risk premia vary with macro conditions; combining penalised estimation with economically constrained representation learning to separate alpha signals that survive those constraints from signals driven by weak factors or data snooping.</li><li>• Research and practice: using the quantitative platform of the Imperial Student Investment Fund to test research signals as implementable strategies, net of transaction costs and within turnover and risk limits.</li></ul> |
| Aug 2026–present | <p><b>AI and Machine-Learning Adviser</b> (part-time), Cost of Capital team, Infrastructure Advisory, KPMG LLP (UK)</p> <ul style="list-style-type: none"><li>• Academic advice on machine-learning methods and on the use of asset-pricing models, such as the conditional CAPM and the betting-against-beta (BAB) factor, in independent estimates of the allowed cost of equity for regulated UK utilities (water and energy networks).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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## Education

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| Aug 2019–Aug 2026 | <p><b>Imperial College London</b>, Imperial Business School, Department of Finance, London</p> <p><b>PhD in Finance</b>, conferred August 2026</p> <p>Supervisors: Prof. Paolo Zaffaroni (main) and Prof. Pasquale Della Corte</p> <p>Thesis: No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing. Examined March 2026.</p> <p><b>MRes in Finance</b>, with Distinction, 2020</p> |
| Sep 2017–Feb 2019 | <p><b>Columbia University</b>, Graduate School of Arts and Sciences, New York</p> <p><b>M.A. in Economics</b> (Financial Econometrics; STEM-designated programme)</p> <p>Adviser: Prof. Jushan Bai. Master's thesis: Diffusion Index Forecasting: A Robust Estimator for Common Shock</p>                                                                                                      |
| Sep 2014–Apr 2017 | <p><b>University of Michigan</b>, Ann Arbor</p> <p><b>B.S. in Economics and Mathematics</b></p> <p>Track: Mathematics of Finance and Risk Management</p>                                                                                                                                                                                                                                       |

## Research

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### Job Market Paper

#### Characteristic Libraries and Portfolio Decisions

Working paper, September 2026

Can a library change that adds no information alter a tuned investment rule? On 153 U.S. characteristics, theme balancing moves target holdings by 19–28% of average gross exposure across four ridge-type procedures and shifts theme attribution, with cost-dependent payoffs; duplicated characteristics act similarly. The 39-to-153 expansion has small, imprecise predictive effects under the evaluated procedures.

PDF

### Working Papers

#### Characteristic-Space Metrics in Factor Models: Identification and Inference

SSRN preprint, September 2026. DOI: 10.2139/ssrn.7445120

Factor models with identical fit can attribute returns differently to latent factors, observed factors and an intercept. The inner product chosen for characteristic space selects this split; the paper formulates the estimand as a loading class modulo rotations and derives chart-invariant inference carrying the Gram metric's estimation error, and drift-robust intervals near an exposure bound.

PDF · Online supplement · SSRN

#### A Geometric Framework for Identification in Characteristic-Based Factor Models

SSRN preprint, March 2026 (preliminary version). DOI: 10.2139/ssrn.7013178. Grows out of the PhD thesis; its inference theory is developed further in Characteristic-Space Metrics.

Which directions in characteristic space are economically distinguishable? The paper takes the metric from the long-run cross-sectional distribution of firm characteristics, states identification and no-arbitrage restrictions as intrinsic geometric properties, and derives No-Arbitrage IPCA (NA-IPCA), which nests IPCA and admits rotation-invariant inference. In geometry-calibrated simulations, Euclidean representations become unstable in anisotropic characteristic spaces.

PDF · SSRN

#### Characteristic Geometry and Portfolio Choice

Working paper, September 2026. Slides available; manuscript in preparation.

Why do factor models with nearly identical forecasts imply different portfolios? On 132 U.S. characteristics, a fixed factor-variance ridge penalises exposures differently across loading representations; removing only that ridge shrinks the largest relative weight gap to about 0.01%. Within one NA-IPCA fit, using the intercept raises gross performance; the post-2007 mean-return gain is imprecisely estimated.

Slides (PDF)

#### Interpreting Estimated Pricing Errors: Evidence from Characteristic-Based Return Forecasts

Working paper, September 2026

What does a smaller estimated pricing error tell a model user? Using 153 U.S. characteristics, the paper separates changes in fitted expected returns from changes in factor loading spaces: the gap in fitted characteristic-intercept magnitudes between equal- and value-weighted training mainly reflects different estimated means, and residual diagnostics add little to forecast selection (exploratory evidence).

PDF

### PhD Thesis

#### No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing

PhD thesis, Imperial College London; degree conferred August 2026. Thesis available on request.

Extends IPCA, via the empirical characteristic metric, to nested QZ-IPCA and NA-IPCA models separating latent, observable (Fama–French five factors plus momentum) and residual-pricing components under a no-arbitrage bound; gives identification results, an asymptotic linear expansion and projected alternating least squares; simulations and a Jensen–Kelly–Pedersen U.S. panel show that similar fit can hide different pricing directions.

## Research Experience

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Aug 2019–Aug 2026    **Doctoral research**, Department of Finance, Imperial College London (asset pricing, financial econometrics, machine learning in finance)

- Models and geometry: characteristic-based factor models with an explicit characteristic-space geometry, using the empirical Gram metric of firm characteristics as the identification convention; the extension of IPCA to the nested QZ-IPCA and NA-IPCA models developed in the thesis.

- Identification and inference: rotation-invariant identification (loading classes modulo latent rotations); asymptotic linear representations that carry an estimated metric's error into standard errors; drift-robust confidence intervals near an exposure bound; projected alternating least-squares estimation; geometry-calibrated Monte Carlo designs.
- Portfolio experiments and trading costs: rolling out-of-sample portfolio construction (120-month windows, U.S. stocks 1973–2024) comparing models under a common risk adjustment; information-preserving library interventions (exact copies, theme balancing) across four tuned ridge-type procedures, traced to theme attribution, stock-level ledgers and returns under transaction-cost scenarios; a separate 39-to-153 characteristic expansion evaluated for stock-return prediction; forecast-evaluation designs that cross expected-return estimates with factor spaces.
- Data and tools: the Jensen–Kelly–Pedersen library of 153 U.S. stock characteristics (jkpfactors.com; WRDS contrib.global\_factor, built from CRSP/Compustat), 1962–2025, with point-in-time formation alignment; a CRSP/Compustat panel built following the Kelly–Pruitt–Su replication dataset (36 characteristics plus a constant, 12,813 firms, 1964–2014); CRSP daily volume for liquidity; NBER recession dates. Author of naipca, a Python toolkit implementing IPCA, QZ-IPCA and NA-IPCA with a companion metric-diagnostics package (available on request).

May 2018–May 2019 **Research Assistant (macro forecasting)**, Department of Economics, Columbia University

- Diffusion-index and factor models that extract common inflation and interest-rate factors from large mixed-frequency macro data; robust PCA and stable principal component pursuit for missing data, revisions, outliers and structural breaks.

Jun 2016–Jan 2017 **Research Assistant (derivatives pricing)**, Department of Mathematics, University of Michigan

- Real-option valuation of a gold mine, using the Lihir Gold Limited IPO as a case: estimated gold-price volatility and the present value of total costs, took strike parameters from the prospectus, and priced the option with a binomial tree.

## Teaching

2023, 2024 **Graduate Teaching Assistant**, Imperial Business School  
Financial Statistics, MSc Risk Management & Financial Engineering. Student rating 4.8 out of 5.  
Nominated by students for the Excellence in Teaching Award (both years).

## Scholarships and Honours

2019–2025 Fully funded doctoral scholarship, Imperial Business School (Graduate Teaching Assistant Scholarship)

2014–2016 University Honors, University of Michigan (three terms)

## Skills and Languages

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| Programming | Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, PyTorch), MATLAB, R, SQL, Git                                                                                                                                                                                                                                                                                 |
| Methods     | Cross-sectional asset pricing (IPCA, QZ-IPCA, NA-IPCA, Fama–MacBeth), latent-factor models, regularised regression and portfolio shrinkage (ridge-type rules, UPSA), robust PCA (RPCA, SPCP), moving-block bootstrap inference, Monte Carlo simulation design, neural networks and transfer learning, state-space and regime-switching models, financial text analysis |
| Data        | CRSP/Compustat and the Jensen–Kelly–Pedersen characteristic library (jkpfactors.com) via WRDS; Bloomberg; LSEG Workspace                                                                                                                                                                                                                                               |
| Languages   | Chinese (Mandarin, native); English (fluent; working language)                                                                                                                                                                                                                                                                                                         |

## References

Available on request.

Updated September 2026